CHARACTERIZATIONS OF INJECTIVE, SURJECTIVE AND BIJECTIVE MORPHISMS OF AN ABELIAN CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the three equivalences; the sentence 'The proof of (2) is similar.' refers to the argument just written out, of which the second assertion is the dual, and it is preserved verbatim.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Kernels as representing objects. Same chapter, Definition definition-kernel and the description of a kernel by the functor it represents. Supplies the description of the kernel of f as the object representing the functor sending z to the kernel of the induced map on Hom-sets, which converts the vanishing of the kernel into injectivity on Hom-sets.
- Coimage and image. Same chapter, the definitions of the coimage and the image of a morphism. Supplies the identifications of the source with the coimage and of the target with the image of a morphism that is both injective and surjective.

Source reference. Original definition 0106: sources/stacks-homology/pinned-original-source.tex, source lines 427–427. Navigation pointer invoked at proof line 883; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : x \to y$ be a morphism in an abelian category $\mathcal{A}$. Then*

(1) *f is injective if and only if f is a monomorphism,*
(2) *f is surjective if and only if f is an epimorphism, and*

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(3) *f is an isomorphism if and only if f is injective and surjective.*

Proof. Proof of (1). Recall that $\mathrm{Ker}(f)$ is an object representing the functor sending z to $\mathrm{Ker}(\mathrm{Mor}_{\mathcal{A}}(z, x) \to \mathrm{Mor}_{\mathcal{A}}(z, y))$, see Definition 0106. Thus $\mathrm{Ker}(f)$ is 0 if and only if $\mathrm{Mor}_{\mathcal{A}}(z, x) \to \mathrm{Mor}_{\mathcal{A}}(z, y)$ is injective for all z if and only if f is a monomorphism. The proof of (2) is similar. For the proof of (3) note that an isomorphism is both a monomorphism and epimorphism, which by (1), (2) proves f is injective and surjective. If f is both injective and surjective, then $x = \mathrm{Coim}(f)$ and $y = \mathrm{Im}(f)$ whence f is an isomorphism. □